Lab 1: Basic statistics in Python

Moments · MoM · MLE · Inference

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Goals

- Compute empirical moments and compare with theoretical values.
- Implement Method of Moments and MLE for the Gaussian.
- Study the sampling distribution of the sample mean by Monte-Carlo simulation.
- Build a confidence interval and run a t-test on a real dataset (Atlanta monthly temperatures).

Setup

- Python 3, dependencies: `numpy`, `scipy`, `matplotlib`.
- Reproducibility: `numpy.random.seed(42)`.
- Dataset: `data/AvTempAtlanta.txt` — monthly average temperatures, Atlanta, USA.
- The skeleton `seed.py` contains function signatures and the visualisation utilities.

Exercise 1 – Empirical moments

(1) Generate $n = 10\,000$ samples from $\mathcal{N}(2, 4)$ (mean 2, variance 4). Compute the four empirical moments:

$$\widehat{\mu}, \quad \widehat{\sigma}^2, \quad \widehat{S} = \frac{1}{n\,\widehat{\sigma}^3}\sum(X_i - \widehat{\mu})^3, \quad \widehat{K} = \frac{1}{n\,\widehat{\sigma}^4}\sum(X_i - \widehat{\mu})^4.$$

Compare with theoretical values $(\mu, \sigma^2, S, K) = (2, 4, 0, 3)$.
(2) Repeat for $n = 100$. Plot $\widehat{\mu}$ as a function of n for $n \in \{10, 50, 100, 500, 1000, 5000, 10\,000\}$. What do you observe?

Exercise 2 – Method of Moments vs MLE for the Gaussian

(1) Implement `mom_gaussian(x)` and `mle_gaussian(x)` returning the estimated $(\widehat{\mu}, \widehat{\sigma}^2)$. Verify numerically that the two estimators *coincide* on the same sample.
(2) Compare the MLE/MoM estimator $\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum(X_i - \bar{X})^2$ with the unbiased one $S_n^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$. Run $B = 1000$ Monte-Carlo replicates of size $n = 20$, draw the histograms of both estimators, and verify empirically that

$$\mathbb{E}[\widehat{\sigma}^2_{\mathrm{MLE}}] = \frac{n-1}{n}\,\sigma^2, \qquad \mathbb{E}[S_n^2] = \sigma^2.$$

Exercise 3 – Sampling distribution of $\bar{X}_n$

(1) For $n \in \{10, 30, 100, 1000\}$, simulate $B = 1000$ independent samples $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2) = \mathcal{N}(2, 4)$, and compute the sample mean of each.
(2) Plot four histograms (one per n) of the B values of $\bar{X}_n$. Overlay the theoretical density of $\mathcal{N}(\mu, \sigma^2/n)$. Compute the empirical variance and compare with σ^2/n.
(3) Repeat with a non-Gaussian distribution (Exp($\lambda = 1$), mean and variance both equal to 1). What does the CLT predict, and is it observed?

Exercise 4 – Confidence interval and t-test on Atlanta temperatures

Load `data/AvTempAtlanta.txt` (one column of monthly temperatures, n values).
(1) Compute $\bar{X}_n$ and S_n. Build the 95% t-confidence interval for μ.
(2) Test $H_0 : \mu = 60$ versus $H_1 : \mu \neq 60$ at level 5%. Report the test statistic, the critical value, the p-value, and the conclusion.
(3) Compare with `scipy.stats.ttest_1samp`.

Exercise 5 – Bonus: bootstrap CI

(1) Implement a non-parametric bootstrap CI for μ (resample the data $B = 5000$ times with replacement, compute $\bar{X}^*$ on each resample, take the empirical $(\alpha/2, 1 - \alpha/2)$-quantiles of the bootstrap distribution).
(2) Compare the bootstrap CI with the t-CI on the Atlanta data. Discuss when each is preferable.

Warning

Always set the seed (`np.random.seed(42)`) before any random draw. Save figures to PNG with explicit names (`lab1_ex3_sampling.png`, etc.).